Essential Docker Concepts for Development

1. Docker Images

- Docker Images are **templates/blueprints**
- They define:
  - What the container contains.
  - The dependencies required by the container.
  - **Images** are the base on which **containers are built.**
  - Image = Blueprint for a container

2. Docker Containers

- Containers are:
  - **Lightweight**
  - **Isolated**
  - **Runtime Environments**
- They are created using Docker Images
- They provide an :
  - **Isolated execution environment**
  - **Reproducible execution environment**
- Applications run inside containers.
- Container = Running Environment created from an Image.

3. Docker Engine

- Docker Engine is the **runtime that manages containers**
- It provides the tools and infrastructure required to
  - Build containers
  - Run containers
  - Manage containers
- Docker Engine = **Infrastructure/runtime used to build, run and manage containers.**

4. Docker File

- A **Docker File** contains instruction for building a **Docker Image.**
- Think of it as the **recipe/instructions for creating the image.**
- It specifies things such as
  - Base Image
  - Application Code
  - Dependencies
  - Configuration setting

Basic relationship:

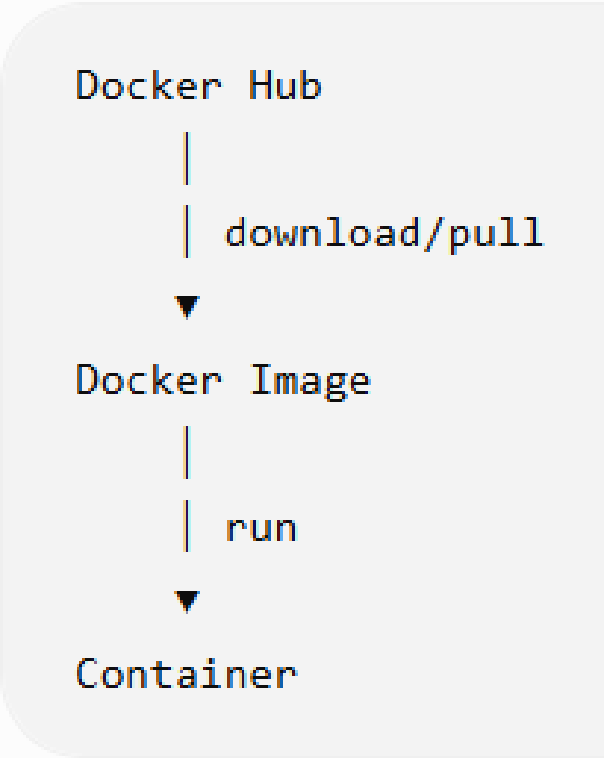
```
Dockerfile
    ↓
  Build
    ↓
Docker Image
    ↓
  Run
    ↓
Container
```

> **Dockerfile = Instructions for building an image**

5. Docker Hub

- **Docker Hub** is a cloud-based registry
- It hosts a large collection of **Docker Images.**
- You can use Docker Hub to
  - Access public images
  - Download Images
  - Share Images
  - Store your own Images
  - Share your images with others

Simple flow:



6. Interview-Ready Summary

- A **Dockerfile** contains **instructions** to build a **Docker Image**.
- An Image acts as a **blueprint** for creating containers
- Containers are lightweight and isolated **runtime environments** in which applications run.
- Docker Engine provides the **infrastructure** to build, run and manage containers
- While Docker Hub is a **cloud-based registry** used to
  - Store
  - Access
  - Download
  - And share Docker Images